1. Clinical Question & Core Hypothesis

Clinical Question

From the Ontologist synthesis, MSI/dMMR should render bladder tumors more immunogenic and thus more responsive to immune checkpoint inhibitors (ICIs), yet immune-architectural imaging features (e.g., TLS/TIL) do not uniformly predict benefit. Why does local immune organization (TLS) sometimes fail to translate into ICI sensitivity in tumors that are expected to be immunogenic?

Working Hypothesis

TLS-like lymphoid aggregates in bladder tumors enable productive anti-tumor immunity only when the tumor harbors a dMMR/MSI-driven high-quality neoantigen landscape, with MSH6 shaping the neoantigen repertoire to support TLS maturation and local T cell activation. Clinically, TLS-high tumors with intact MSH6 in the context of dMMR/MSI will show higher therapy_response to immune checkpoint inhibitors, whereas TLS-high tumors with MSH6 deficiency or non-MSI backgrounds will exhibit reduced response due to TLS dysfunction and immune exhaustion, implying a need for combination strategies in those cases.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	dMMR/MSI, MSH6
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

dMMR/MSI: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

MSH6: MSH6 as a specific MMR component with prognostic and mechanistic importance (it influences the quality and spectrum of neoantigens).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	CTCs	Supporting modifier from knowledge graph
Secondary	GREM1	Supporting modifier from knowledge graph

Secondary	TGF-β	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

Prediction 2: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Multi-modal validation pipeline: computational -> spatial -> functional, providing mechanistic depth beyond association.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- CTCs: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- GREM1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.